DELFT UNIVERSITY OF TECHNOLOGY
Faculty of Electrical Engineering, Mathematics and Computer Science

CSE2315 Lab Course
Solutions 5

deadline: March 12, 2026, 13:45

1. Consider the alphabet $\Sigma = \{a, b, c\}$ and the language

$$L = \{a^k b^k c^k \mid k \geq 1\}.$$

Give a high-level description for a three-tape enumerator that enumerates L. You may want to use the second and third tapes as counters. You do not have to give implementation details of the counters.

Solution: The following enumerator M enumerates L:

"On empty input:

1. Write 1 on the third tape.
2. Write 0 on the second tape.
3. Clear the first tape and move the head to the very left.
4. Write an a on the first tape and move the head to the right.
5. Increment the counter on the second tape.
6. If this counter is less than the counter on the third tape, go to to step 4.
7. Clear the second tape and write a 0.
8. Write a b on the first tape and move the head to the right.
9. Increment the counter of the second tape.
10. If this counter is less than the counter on the third tape, go to step 8.
11. Write a c on the first tape and move the head to the right.
12. Decrement the counter of the second tape.
13. If this counter is more than zero, go to step 11.
14. At this point, a new word in L is on the first tape. Increment the counter on the third tape and go to step 2."

2. *Old exam question endterm 23-24:*

(a) Give a Turing Machine M that is not a decider such that $L(M)$ is decidable.

Solution: Lot of possibilities.

(b) Explain why M is not a decider.

Solution: Depends on M.

(c) Explain why $L(M)$ is decidable.

Solution: Depends on M.

3. *Old exam question resit 20-21:* Let $MIA_{TM} = \{\langle M_1, M_2 \rangle \mid L(M_1) \cap L(M_2) \neq \emptyset\}$. Show that this is a recognizable language.

Solution: We can try all the words and run M_1 and M_2 on them, using dovetailing to consider i steps of all words with length $< i$ in each iteration. If one of the words is accepted by M_1 and M_2, then indeed there exists a word in $L(M_1) \cap L(M_2)$ and thus we will also accept. (Note that this is no decider, as M could loop on one of the words without any of the words accepting, in which case our machine also loops.)

4. In the game of chess, consider an intermediate board configuration, and suppose it's white's move. We know that white is either guaranteed to win or not. Let

$$L = \{w \mid w \text{ represents a board configuration, and white is guaranteed to win if it is white's move and white plays optimally}\}.$$

Is L recognizable, decidable, or neither? Explain.

Solution: L is decidable because it is finite. We can create a TM with all words in L hardcoded in its state machine. Coming up with an explicit description for such a TM is practically difficult, but that's another story. We do know there exists some TM that decides L, so L is decidable.

5. *Revision, Old exam question, midterm 21-22:* Given a DTM M with $\Sigma = \{a, b\}$ such that:

- M has no transition to its q_{reject} i.e. the reject state is an unreachable state
- $L(M) = \{w \in \Sigma^* \mid w \text{ starts with an } a\}$

Is M a decider, not a decider, or can we not conclude that? Explain your answer.

Solution: The fact that $L(M)$ is decidable does not imply that M is a decider. On the contrary, since the reject state can never be reached and the computation path can also not stop in a deterministic turing machine, M loops on all words $w \notin L(M)$. Since $L(M) \neq \Sigma^*$, at least one such word exists. That means M loops on at least one word, making M not a decider.

6. *Old exam question resit 20-21:*

In a log of a scheduler, it is noted down when a request for a task is given, and when an action is done to do that task. Requests are noted down with r, actions with a. A log is a final log if all tasks have been done. Thus, a final log can be described by the language of words where every r has a corresponding a that appears later in the string, formally given as $L = \{w \mid w$ has an equal amount of r's and a's and the $i^{th} a$ appears later in the string than the $i^{th} r$, for each i in $(1, |w|/2)\}$. Give an informal description of a Turing machine with the alphabet $\{r, a\}$ that decides this language. Use a maximum of 15 lines.

As an example: $rarraa \in L$ and $raaarr \notin L$

Solution: $M =$ "On input w:

1 Find the first non-blank symbol

2 If there is only blank symbols, accept.

3 If it's a a, reject.

4 Cross it off.

5 Find the first a.

6 If it isn't present, reject.

7 Cross it off.

8 Return to the front of the tape and go to stage 1.

7. *Revision, old exam question midterm 19-20:* Consider the following rules R of a CFG $G = (V, \Sigma, R, S)$, with $V = \{S, A, B\}$:

$$\begin{aligned} S &\to ASA \mid Ba \\ A &\to bB \mid S \\ B &\to c \mid b \mid \varepsilon \end{aligned}$$

(a) (1 point) Give a valid set Σ such that $|\Sigma| = 5$.

Solution: $\{a, b, c\} \subseteq \Sigma$. For example $\{a, b, c, d, e\}$. Note that $V \cap \Sigma = \emptyset$ and $\varepsilon \notin \Sigma$ must both hold.

(b) (6 points) Convert G into Chomsky Normal Form (CNF). Use the procedure from Sipser, showing intermediate steps.

Solution: Introduce a new start variable:

$$\begin{aligned} S_0 &\to S \\ S &\to ASA \mid Ba \\ A &\to bB \mid S \\ B &\to c \mid b \mid \varepsilon \end{aligned}$$

Remove ε-rules:

$$\begin{aligned} S_0 &\to S \\ S &\to ASA \mid Ba \mid a \\ A &\to bB \mid S \mid b \\ B &\to c \mid b \end{aligned}$$

Remove unit rules:

$$\begin{aligned} S_0 &\to ASA \mid Ba \mid a \\ S &\to ASA \mid Ba \mid a \\ A &\to bB \mid ASA \mid Ba \mid a \mid b \\ B &\to c \mid b \end{aligned}$$

Convert remaining rules:

$$\begin{aligned} S_0 &\to U_sA \mid BU_a \mid a \\ S &\to U_sA \mid BU_a \mid a \\ A &\to U_bB \mid U_sA \mid BU_a \mid a \mid b \\ B &\to c \mid b \\ U_s &\to AS \\ U_a &\to a \\ U_b &\to b \end{aligned}$$

Sorry, no optional extra ~~exercise~~ puzzle this week. But are you up to date with the Quadruple Quest?

End of the assignment